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Mastering SQL Joins: A Practical Guide for beginners.

#datascience #sql

Introduction

SQL joins allow you to combine data from multiple tables based on relationships between them. In this article, we'll explore different types of joins using this two schemas:

- **City Hospital Schema**
- **Nairobi Academy Schema**

What is a SQL Join?

A **join** retrieves data from two or more tables using a related column.

Basic Syntax

```
SELECT columns
FROM table1
JOIN table2
ON table1.column = table2.column;
```

1. INNER JOIN

Returns only records that exist in both tables.

Example (City Hospital schema)

```
SELECT p.full_name AS patient_name,
    d.full_name AS doctor_name,
    a.appointment_date,
    a.diagnosis
FROM appointments a
INNER JOIN patients p
    ON a.patient_id = p.patient_id
INNER JOIN doctors d
    ON a.doctor_id = d.doctor_id;
```

Only matching records appear while missing relationships are excluded

2. LEFT JOIN

Returns all records from the left table and matches from the right.
Example

```
SELECT
    p.full_name,a.appointment_date,a.diagnosis
FROM patients p
LEFT JOIN appointments a
    ON p.patient_id = a.patient_id;
```

All patients are shown while missing appointments appear as NULL
This shows all primary records, even if incomplete

3. RIGHT JOIN

Returns all records from the right table.

Example

```
SELECT d.full_name,a.appointment_date,a.diagnosis
FROM appointments a
RIGHT JOIN doctors d
    ON a.doctor_id = d.doctor_id;
```

All doctors appear while missing appointments show NULL
The Focus on the right-side table.

4. FULL OUTER JOIN

Returns all records from both tables, matched or not.

```
SELECT
    p.full_name,
    a.appointment_date
FROM patients p
FULL OUTER JOIN appointments a
    ON p.patient_id = a.patient_id;
```

Combines LEFT + RIGHT JOIN behavior
You will see everything, including gaps.

5. CROSS JOIN

Returns all possible combinations.

Example (Nairobi Academy)

```
SELECT
    s.student_name,
    e.exam_name
FROM students s
CROSS JOIN exams e;
```

Produces Cartesian product
This Generates combinations

6. SELF JOIN

A table joins with itself.
Example

```
SELECT
    s1.student_name AS student_1,
    s2.student_name AS student_2,
    s1.class
FROM students s1
JOIN students s2
    ON s1.class = s2.class
    AND s1.student_id <> s2.student_id;
```

Useful for comparisons or hierarchies
This Compares rows within the same table.

7. LEFT ANTI JOIN (Find Missing Data)

Returns records in the left table with no match.

```
SELECT p.full_name
FROM patients p
WHERE NOT EXISTS (
    SELECT 1
    FROM appointments a
    WHERE a.patient_id = p.patient_id
);
```

This finds missing relationships.

8. RIGHT ANTI JOIN

```
SELECT d.full_name
FROM doctors d
WHERE NOT EXISTS (
    SELECT 1
    FROM appointments a
    WHERE a.doctor_id = d.doctor_id
);
```

Finds unused or inactive records

Nairobi Academy: **Combining Data**
Student Exam Results

```
SELECT s.student_name,e.exam_name,r.marks
FROM results r
```

```
INNER JOIN students s
    ON r.student_id = s.student_id
INNER JOIN exams e
    ON r.exam_id = e.exam_id;
```

Ranking Students

```
SELECT s.student_name,e.exam_name,r.marks,
    RANK() OVER (PARTITION BY e.exam_name ORDER BY r.marks DESC) AS rank
FROM results r
JOIN students s ON r.student_id = s.student_id
JOIN exams e ON r.exam_id = e.exam_id;
```

Here is a shortcut to choosing the Right Join

*Join Type *	*When to Use *
INNER JOIN	Only matching data
LEFT JOIN	All left + matches
RIGHT JOIN	All right + matches
FULL OUTER JOIN	Everything
CROSS JOIN	All combinations
SELF JOIN	Compare within table
ANTI JOIN	Find missing data

Conclusion

SQL joins are the foundation of relational data analysis.
Mastering them allows you to:

- Combine datasets effectively
- Identify missing or inconsistent data
- Build powerful reports

The real skill isn't just writing joins,it's understanding when to use each one.

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